

VDJ: Visualization for T- and B-cell repertoires

This guide describes VDJ, a web-based tool for visualizing and interpreting the results of immunosequencing assays (e.g., those marketed by [Adaptive Biotechnologies](#)).

NOTE: VDJ is not provided by or associated with Adaptive Biotechnologies. The code for VDJ is open-source, [available on GitHub under MIT license](#). Details on modifying, building and hosting the application are available in the GitHub repository.

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Accessing the VDJ Tool

The simplest way to use VDJ is to download and run it locally on your laptop or desktop.

[Prerequisites and instructions for doing this can be found here.](#)

Alternatively, your institution or immunosequencing vendor may “host” a shared version of the tool for their stakeholders. If so, they will provide you with a URL and login credentials.

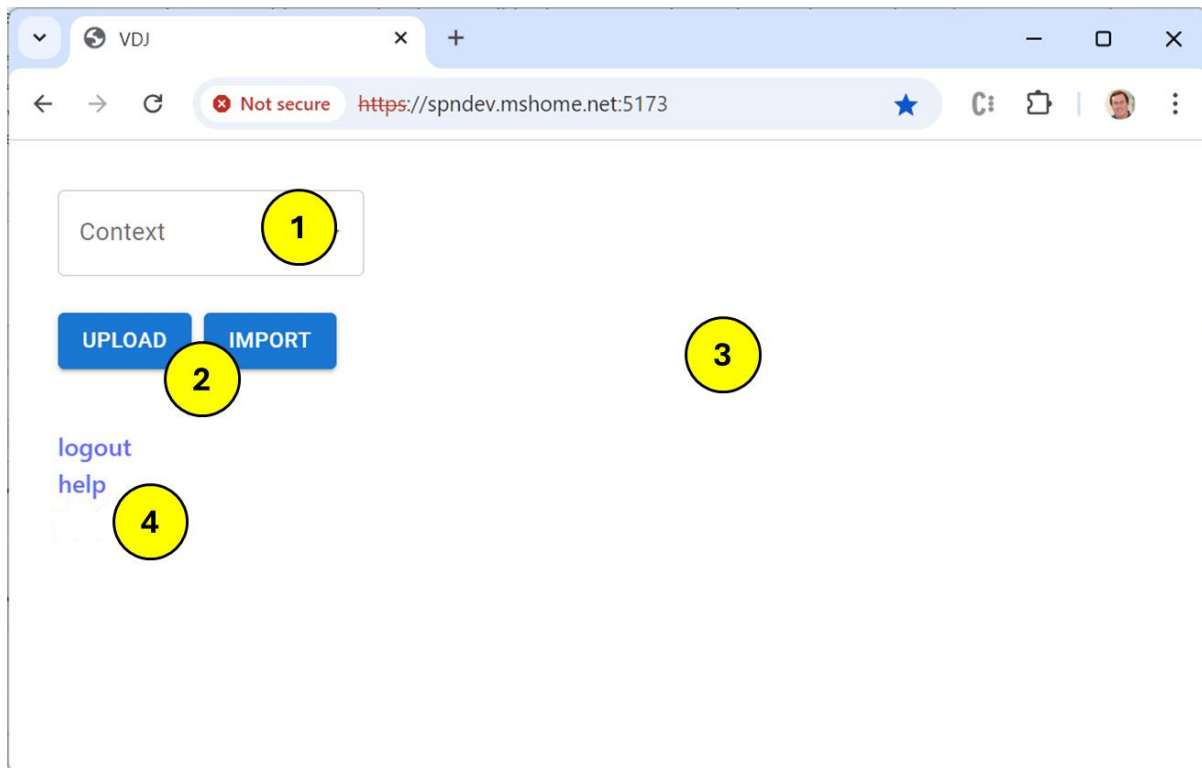
Lastly, you can build, host, and even modify or improve a version of VDJ yourself. This option requires some technical knowledge; [details to get you started can be found here.](#)

Overview and Navigation

The main VDJ interface consists of a left-side navigation/action bar and a right-side content area. Related “repertoires” (samples) are grouped together into “contexts” --- a context may represent a patient, a study, or any group of samples that are analyzed together.

Note: by default each user will see their own collection of contexts and repertoires. If a group of users regularly collaborates on the same data, the VDJ administrator can assign those accounts to a “group” so that data is shared.

On initial login, no context is “active” and the screen will appear like this:



1. Click the context menu to see a list of available contexts and select one to work with. You can also type into this box to search for a context rather than scroll through them all.
2. The “Upload” and “Import” buttons are always available; this is how repertoires are added to your data set. They are described in the “Managing Repertoires” section.
3. The right-hand side of the application starts out empty. This is where content will appear as you begin to work with repertoires.
4. The “logout” button terminates your working session; “help” will open the user manual in a new tab.

Once a context is selected using the dropdown, the repertoires in that context will be shown on the navigation bar and new action buttons will become available:

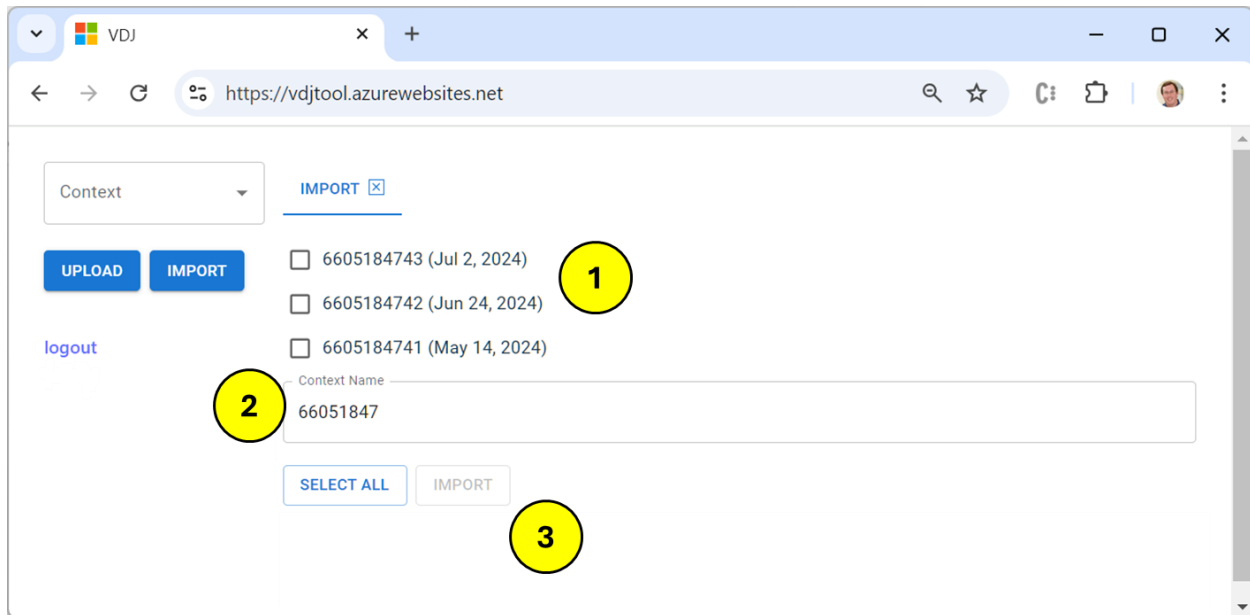
The screenshot shows a web browser window with the URL <https://spndev.mshome.net:5173>. The page has a header with a 'Context' dropdown and an 'UPLOAD' button with a close icon. Below the header, there are two main buttons: 'UPLOAD' and 'IMPORT'. To the left of these buttons are links for 'logout' and 'help'. The main form area contains several input fields: a 'Context Name' field (1), a 'Repertoire Name' field, a 'Collection / Sample Date (Optional)' field with a date picker icon (3), a 'Choose File' button (2) with the text 'No file chosen', and a 'GO' button (4). The 'GO' button is disabled.

1. Enter the context in which the repertoire should be saved. This context does not have to already exist; new ones will be created automatically.
2. Choose a file on your local machine to upload. These files may end in “.tsv” or they may be compressed (“.tsv.gz” or “.tsv.zip”). Using a compressed file is recommended; doing so can dramatically reduce upload time. Choosing a file fills in the “Repertoire Name” field but you may update it if desired.
3. Optionally, provide a collection date for the sample. This value is used for display and sorting in some visualizations, but is not required.
4. Click the “Go” button to start the upload. After the upload is complete, close the tab using the “X” at the top of the content pane.

If this is the first repertoire in a new context, refresh your browser so it appears in the “context” dropdown. If the context already exists, the new repertoire will show up automatically.

Import from Agate

Adaptive stores customer data in a cloud-based system called “Agate.” Agate is not part of VDJ, but repertoires stored there can be imported into the tool. Begin an import by clicking the “Import” button, following the login instructions (if necessary), and providing a search string. The search is case-insensitive and searches within sample, project and order names. This query can take a bit of time to show results:



1. Check the boxes for samples to be imported. To select all listed samples, use the “Select All” button.
2. Verify the context for samples. VDJ attempts to pick a reasonable context name, but you can use whatever name works best for you.
3. Click “Import” to start the process; the resulting page will show progress.

Note: If you encounter an error during import, try logging out and logging back in again. This will refresh your access token and may resolve the problem.

If a selected repertoire already exists in the desired context, it will be skipped. This makes it easier to update data for a context over time. Just repeat a search, “select all” and then import --- new repertoires will be added and existing ones ignored.

As with Update, if this is the first repertoire in a new context, refresh your browser so it appears in the “context” dropdown. If the context already exists, new repertoires will show up automatically.

Delete

The “Delete” button can be used to remove repertoires present in the current context.

When the last repertoire is removed from a context, that context is automatically removed as well. You can recreate it at any time by Uploading or Importing repertoires using the original context name.

Actions

Top 100

This is the most common action for viewing details of a single repertoire. To make the feature performant, only the top 100 rearrangements sorted descending by the selected metric (% of Locus by default) are displayed.

Context: demo

☐ A_TCRB_ID

☒ A_TCRB_MRD

TOP 100 DETAILS

SEARCH OVERLAP

UPLOAD IMPORT

DELETE

logout

TOP 100 - A_TCRB_MRD

Count: 106,556 Uniques: 97,411

TCRB: 106,556 Cells: 1,229,340

Sort rearrangements by: % Locus

Locus	Count	% Locus	% Cells	Rearrangement	Amino Acid	V
TCRB	88	0.0826	0.0072	CTGCAGCCAGAAAGACTCGGCCCTGTATCTCTGGCCAGCAGCCTCGACCTCTCCGGGACGGTGACCTATGGCTACACCTTCGGTTGG	CASSLDLSGTVTYGYTF	TCR
TCRB	75	0.0704	0.0061	CTCAGAACCGGGAGACACGGCACTGTATCTCTGGCCAGCAGTCAATCTGGGACACCCCGGGGGTACGAGCACTACTTCGGGGCG		TCR
TCRB	53	0.0497	0.0043	GAGTCGGCTGCTCCCTCCAAACATCTGTGTACTTCTGTGGCAGCAGTTACTATCGAGGGAGCAATCAGCCCGCAGCATTTTGGTGAT	CASSYYRGSNQPHF	TCR
TCRB	51	0.0479	0.0041	GTCCGCCAGCACCACAGACATCTATGTACTCTGTGTATGAGGAGGACAGGGGGCTTCACCCCTCCACTTTGGGAG		TCR
TCRB	37	0.0347	0.0030	CCGCTCAGGCTGCTGTGGCTGCTCCCTCCAGACATCTGTGTACTTCTGTGGCAGCGGATTAGATGAAACTGTTTTCGGCAGT	CASGLDEKLFF	TCR
TCRB	35	0.0328	0.0028	CCTGTGGGGTTGGAGTCGGCTGCTCCCTCCAAACATCTGTGTACTTCTGTGGCAGCAGTTAGGGCTGAAGCTTTCTTTGGACAA		TCR
TCRB	34	0.0319	0.0028	TTGGAGCTGGGGACTCGGCCCTTTATCTTTGGCCAGCAGCTTGGCCAGACAGGGGTCTCTGGAACACCATATATTTTGGAGAG	CASSLAQTGVSNTIYF	TCR
TCRB	34	0.0319	0.0028	CGCACAGACAGGGGGACTCGGCCATGTATCTCTGTGCCAGCAGCGCCCGGGGGTTGGAGCACAGATACCGATATTTTGGCCCA	CASSGPGVSTDTQYF	TCR
TCRB	29	0.0272	0.0024	ACCTTGGAGCTGGGGACTCGGCCCTTTATCTTTGGCCAGCAGCTTGTGGAGTCAGGGGGAACTTACGAGCACTACTTCGGGGCG	CASSLWSQGTYEQYF	TCR

- The header shows aggregate values for the repertoire:
 - Total sequence count
 - Sequence counts grouped by locus group (TCRAD, TCRB, TCRG, IGH, IGKL)
 - Total number of unique sequences
 - Count of cells in the sample, or milliliters for cell-free repertoires.
- The dropdown changes the sort metric:
 - Simple Count
 - Percentage of this sequence within cell count (or milliliters for cell-free)
 - Percentage of this sequence within its locus group, by count
- The results table contains the following columns:
 - Locus (TCRAD, TCRB, TCRG, IGH, DJ, IGK, IGL)
 - Count as estimated by the assay
 - % Locus
 - % Cells (or Count/ML for cell-free)
 - Nucleotide rearrangement sequence.
 - Amino Acid, if the sequence is productive
 - Identified V, D and J genes or alleles (depending on assay confidence)
 - Probability of sequence uniqueness, if available (log10)
 - A link to open an analysis of this rearrangement using [IMGT V-Quest](#).

Details

The details action uses the same display format as Top 100, but returns all rearrangements in the repertoire. The rearrangements are unsorted and appear in the order they are present in the source TSV file. The “FORWARD” and “BACK” links page through the table.

Search

VDJ can search multiple samples for instances of a target nucleotide or amino acid sequence.

The screenshot shows the VDJ web application interface. At the top, there's a browser window with the URL <https://spndev.mshome.net:3001>. The interface includes a search bar with a dropdown menu set to 'demo'. Below the search bar, there are checkboxes for 'B_Cell_ID' and 'B_Cell_MRD'. To the right of these, there are radio buttons for 'Nucleotide', 'CDR3', 'Amino Acid', 'MRD', and 'Genes'. A text input field for the 'Nucleotide Sequence' is highlighted with a red border and a yellow circle labeled '2'. Below this field, there is a checkbox for 'Match full sequence' and a text input for 'Allowed Mutations' set to '0'. A 'GO' button is at the bottom right. On the left side, there are buttons for 'TOP 100', 'DETAILS', 'SEARCH', 'OVERLAP', 'TRACK', 'GENES', 'EXPORT', 'UPLOAD', 'IMPORT', and 'DELETE'. A yellow circle labeled '1' is placed over the 'SEARCH' button. A yellow circle labeled '3' is placed over the 'Match full sequence' checkbox. A yellow circle labeled '4' is placed over the 'Allowed Mutations' input field.

1. Select a target for the search (i.e., the “haystack”):
 - a. Nucleotide: anywhere in the nucleotide sequence
 - b. CDR3: the identified CDR3 region, a subset of the full sequence
 - c. Amino Acid: the translated protein sequence for the CDR3 region
 - d. (See below for notes on MRD and Genes search)
2. Enter the nucleotide or protein sequence to search for (i.e., the “needle”)
3. If “Match full sequence” is checked, the needle and the haystack must match exactly, otherwise the needle may be any substring of the haystack. For example, the amino acid sequence “**CAEENWN**” will match the sequence “**CAEENWN**YGFDPW” only if this box is UNchecked.
4. “Allowed Mutations” accepts a given number of mismatches (i.e., substitutions) between the needle and the haystack. For example, if this value is one, the needle “CAEEN**WN**” will match the haystack “CAEEN**AN**”. This parameter works only for substitutions, not insertions or deletions.

MRD Search

“MRD” searching attempts to enable consistent search results even when different immunosequencing assays have been used to generate data. This search is similar to the one used by the Adaptive [clonoSEQ](#) clinical report, but you should **not** assume it is identical.

In this mode each sequence in the “haystack” is aligned with the “needle” using the J edge of the CDR3 region as an anchor (if this value is not present in the pipeline data, the “right” edge of the sequence is used). Matching proceeds to the left and right of this location as long as there is overlap. A “match” must be at least 25 bases in length.

Genes Search

This version of search filters for sequences generated from specific V, D or J genes. The input is a comma-separated list of gene name substrings; all must be found in order to call a match.

For example, the search string “IGHV06,IGHJ04-01*02” will match all sequences for which the V gene is in the IGHV06 family **and** the J gene is the IGHJ04-01*02 allele.

Overlap

The Overlap action identifies commonality between two or more repertoires (currently set to a maximum of six). Comparison can be done by full nucleotide sequence, identified CDR3, or amino acid sequence.

Context: demo

OVERLAP

Overlapping: B_Cell_ID, B_Cell_MRD

OPEN SCATTER PLOT

TOP 100 DETAILS

SEARCH OVERLAP

UPLOAD IMPORT

DELETE

logout

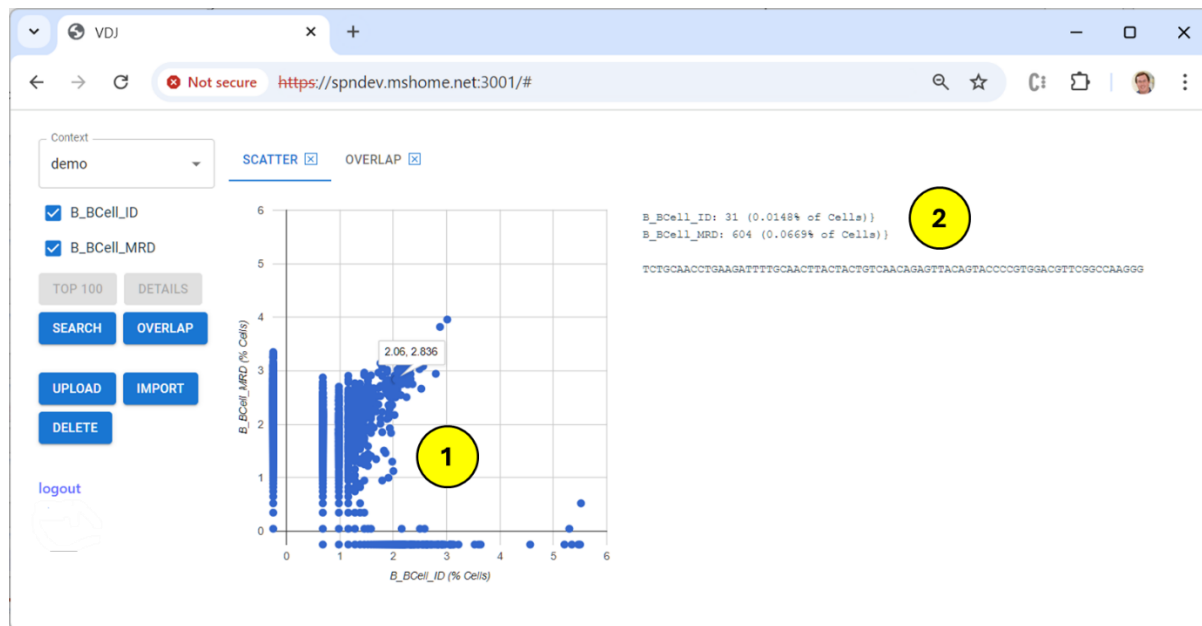
Rearrangement	Present In	Max Count	B_Cell_ID	B_Cell_MRD
ACAGTGGGTATAACTGGAACACGGCTGGTTCGACCCCTGGGGCCAGGGAACC	2	68820	68820	3
ACAGTGGGTATAACTGGAACACGGCTGGTTCGACCCCTGGGGCCAGGGAACC	2	41620	41620	1
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	8185	215	8185
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	5949	157	5949
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	2613	58	2613
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	2544	78	2544
ATCTCTGGGCTCCAGGCTGAGGACGAGGCTGATTACTGTCAGCTCATATAACAGCAGCAGCACTGGGTGTTCCGCGGAGGGACCAAGC	2	2342	158	2342
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	2090	50	2090
CTGACTATTACTGTGACAGCTGGGGCACTGGCATTGGGTGTTCCGCGGAGGGACCAAGC	2	1783	42	1783
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	1654	54	1654
AGGCTGATTATTACTGTGACGATGGGATGACAGCCTGAGTGGTGGGTGTTCCGCGGAGGGACCAAGC	2	1645	79	1645
TCTGCAACCTGAAGATTGTGCACTTACTGTCAACAGAGTTACAGTACCCCTCACTTTCGCGGAGGG	2	1530	48	1530
AGGCTGATTATTACTGTGACGATGGGATGACAGCCTGAAATGGTGGGTGTTCCGCGGAGGGACCAAGC	2	1493	130	1493
GTGGCCGTGGCCACCCCTGTCTGCCCTCTGCCGTAGCCAGCTTCTCTGAGCCCTAGTGGCAGCCAGGG	2	1433	49	1433

1. The resulting table shows the overlapping nucleotide or protein sequence. Note that in the case of “CDR3” matching this is a substring of the full nucleotide sequence. Clicking on a sequence will open a new “Search” tab showing detailed results.
2. Other columns in the table provide information on the overlap:
 - a. “Present In” specifies the number of samples containing the overlap.
 - b. “Max Count” is the maximum count found in any sample.
 - c. Each repertoire is then represented in a column with its count value.

Only sequences that appear in at least two repertoires are returned in this table, which is sorted descending first by “Present In” and then by “Max Count.”

Overlap Scatter Plot

When the Overlap action is run on exactly two repertoires, an additional option “Open Scatter Plot” will be present on the results pane. Clicking this button will open a comparison chart between the two repertoires using the same metric.



The chart uses a log10 scale to plot normalized values. This typically a percentage of cells, but can be count per milliliter for cell-free repertoires). Sequences with equivalent abundance in both repertoires will trend along the SW-NE diagonal line; those with differential abundance will fall in the top-left or bottom right quadrant.

Clicking on a data point will display the sequences it represents including exact count values.

Unlike the overlap results table, ALL sequences in both sequences are represented on the chart, including those that appear in one of the two repertoires. These values are plotted just below and to the left of the axis lines.

Track

This action charts the abundance of dominant rearrangements across a number of related repertoires. It is similar to but not exactly the same as the information presented in MRD clinical reports. See the “MRD Search” section for details on the matching algorithm.

VDJ <https://vdjtool.azurewebsites.net>

Context: test TRACKING

☒ A_BCell_ID
☒ A_BCell_MRD
☐ A_TCRG_MRD
☐ A_TCRG_ID
☐ D_BCell_Cellfree_MRD

TOP 100 DETAILS
 SEARCH OVERLAP
 TRACK EXPORT
 UPLOAD IMPORT
 DELETE

GO

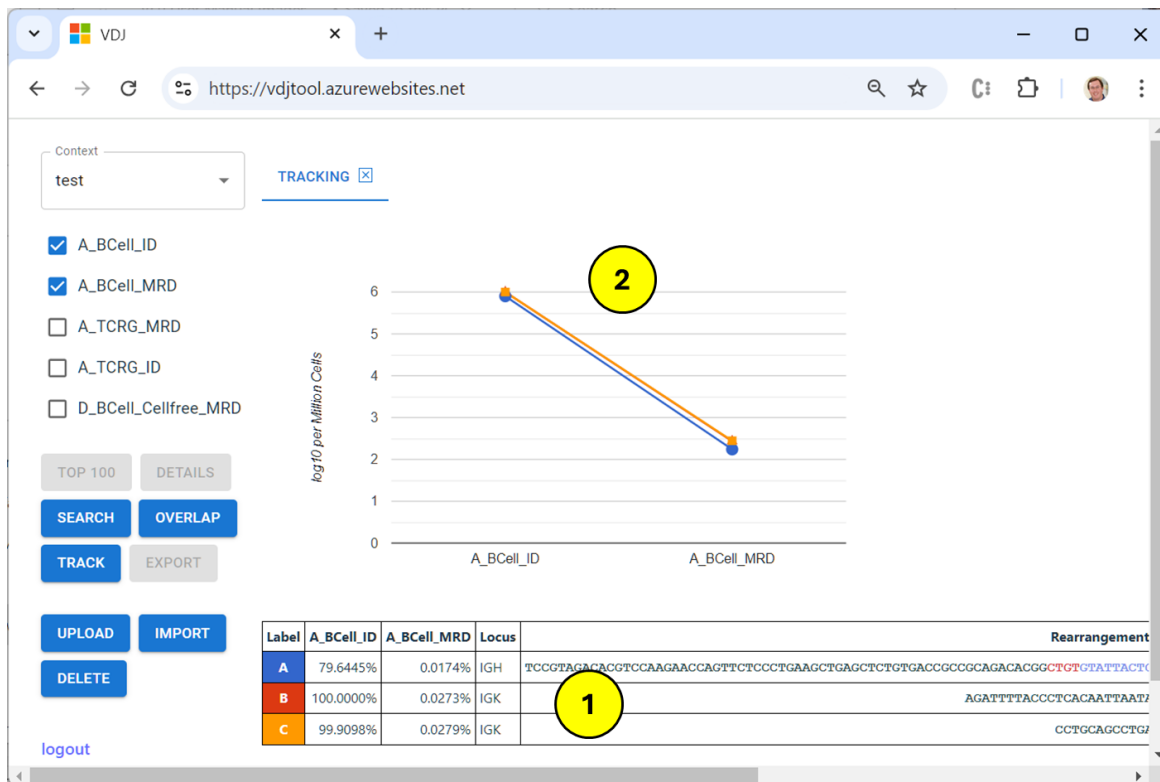
A_BCell_ID

	Locus	Count	% Locus	% Cells	Rearrangement
☒	IGH	98,581	98.5811	79.6445	TCCGTAGACACGTCGAAGAACAGTTCCTCCCTGAAGCTGAGCTCTGTGACCGCCGAGACACGGCTGTATTACTGGAGGAAATATTAGTAGT
☒	IGH	51,432	51.4323	100.0000	AGATTTTACCCTCACAATTAATACATAGAATCTGAGGATGCT
☒	IGK	46,540	47.9403	99.9098	CCTGCAGCCTGAAGATTTTGCACCTTATTACT

A_BCell_MRD

	Locus	Count	% Locus	% Cells	Rearrangement
☐	IGH	5,963	5.9631	0.0266	CTTTGTCTCTGGGAGGACAGGACTGTGTCCCTGTGTGATGCT
☐	IGK	5,685	5.6854	0.0279	CCTGCAGCCTGAAGATTTTGCACCTTATTACTGCCACAGTATATAGTTAC
☐	IGK	285	5.5491	0.0273	AGATTTTACCCTCACAATTAATACATAGAATCTGAGGATGCTGCATATTACTTCTGTCTAG

1. When the track action is selected, sequences are shown from the selected repertoires that are either (a) marked as “Dx” or (b) have abundance $\geq 5\%$ of locus.
2. Unique “Dx” rearrangements (i.e., those identified as dominant in the data) are selected by default, but the checkboxes can be used to follow other rearrangements of interest.
3. Clicking the “Go” button displays the tracking chart and table.

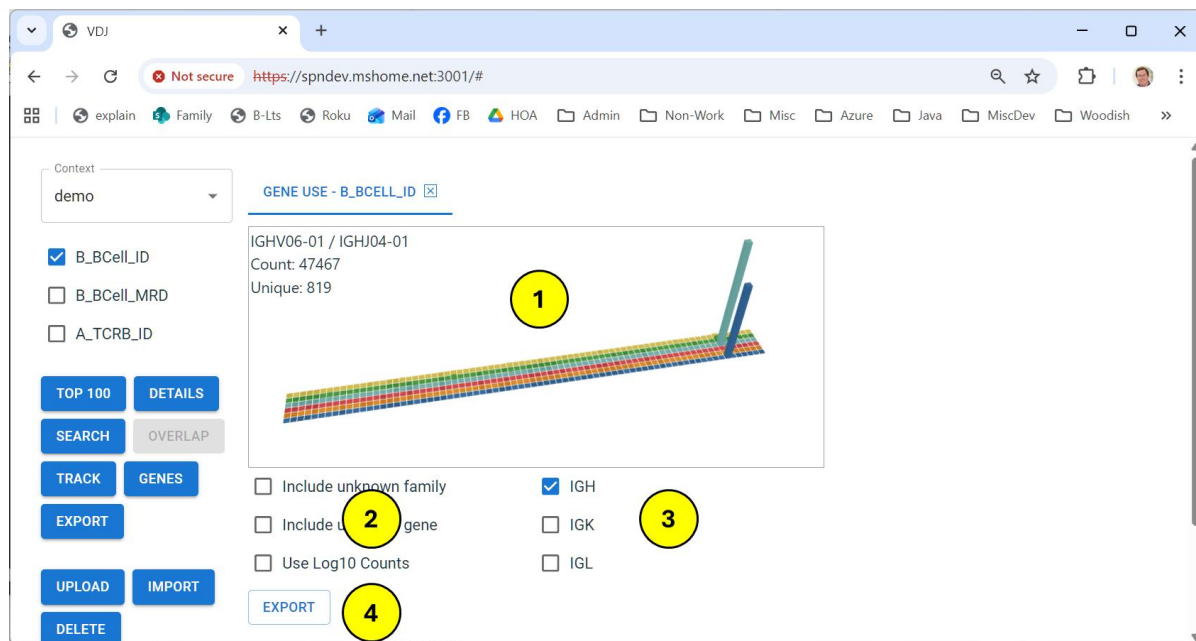


1. The tracking table shows the abundance of each selected rearrangement in each selected repertoire. The color and letter in the “Label” column can be used to correlate this table with the chart above.
2. The tracking chart displays the change of each rearrangement in abundance across the selected repertoires. The chart uses a log10 scale to help visualize large changes in these values.
3. Clicking the “search” link to the far right of this table will open a new tab containing the details of matching sequences.

Gene Use

This action visualizes the use of V and J genes within a single repertoire. V and J genes from the repertoire are represented on the horizontal axes, with bars representing template counts. The view can be manipulated using the mouse:

- Hovering over a bar shows the relevant genes and counts.
- Clicking a bar opens a tab showing details of the relevant sequences.
- Scrolling within the visualization zooms in and out.
- Left-dragging within the visualization moves the viewing angle.
- Right-dragging within the visualization pans the scene.



1. The visualization area as described above.
2. These checkboxes include or exclude sequences where no V family or gene was called, and allow toggling between natural and log10 counts for bar height.
3. Locus checkboxes enable/disable viewing of specific loci within the visualization.
4. The “export” button saves a copy of the current visualization as a PNG graphic.

Export

Use this action to save a repertoire file to your local computer:

1. "TSV" is exactly the file that was added to the VDJ tool by upload or import.
2. "FASTA" is a FASTA-compliant field with one entry per sequence in the repertoire. Two versions are available to assist in mapping output using this file:
 - a. "by Row" sets the description for each sequence to a 0-based row number.
 - b. "by Hash" sets the description for each sequence to the SHA256 value of the sequence itself.